1 The points A, B and C have coordinates A(3, 2, –1), B(–1, 1, 2) and C(10, 5, –5), relative to the origin O.

Show that $\overrightarrow{OC}$ can be written in the form $\lambda\overrightarrow{OA} + \mu\overrightarrow{OB}$, where λ and μ are to be determined.

What can you deduce about the points O, A, B and C from the fact that $\overrightarrow{OC}$ can be expressed as a combination of $\overrightarrow{OA}$ and $\overrightarrow{OB}$?

**[6]**

2

In this question, the unit vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ are in the directions east and north.

Distance is measured in metres and time, *t*, in seconds.

A radio-controlled toy car moves on a flat horizontal surface. A child is standing at the origin and controlling the car.

When $t = 0$, the displacement of the car from the origin is $\begin{pmatrix}0\\-2\end{pmatrix}$ m, and the car has velocity $\begin{pmatrix}2\\0\end{pmatrix}$ $\mathrm{m\,s^{-1}}$.

The acceleration of the car is constant and is $\begin{pmatrix}-1\\1\end{pmatrix}$ $\mathrm{m\,s^{-2}}$.

(i) Find the velocity of the car at time *t* and its speed when $t = 8$.

**[4]**

(ii) Find the distance of the car from the child when $t = 8$.

**[4]**

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3 In this question take $g = 10$.

The directions of the unit vectors $\begin{pmatrix}1\\0\\0\end{pmatrix}$, $\begin{pmatrix}0\\1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\0\\1\end{pmatrix}$ are east, north and vertically upwards.

Forces **p**, **q** and **r** are given by $\mathbf{p} = \begin{pmatrix}-1\\-1\\5\end{pmatrix}$ N, $\mathbf{q} = \begin{pmatrix}-1\\-4\\2\end{pmatrix}$ N and $\mathbf{r} = \begin{pmatrix}2\\5\\0\end{pmatrix}$ N.

(i) Find which of **p**, **q** and **r** has the greatest magnitude. **[2]**

(ii) A particle has mass 0.4 kg. The forces acting on it are **p**, **q**, **r** and its weight.

Find the magnitude of the particle's acceleration and describe the direction of this acceleration. **[4]**

4

In this question the origin is a point on the ground. The directions of the unit vectors $\begin{pmatrix}1\\0\\0\end{pmatrix}$, $\begin{pmatrix}0\\1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\0\\1\end{pmatrix}$ are east, north and vertically upwards.

Alesha does a sky-dive on a day when there is no wind. The dive starts when she steps out of a moving helicopter. The dive ends when she lands gently on the ground.

- During the dive Alesha can reduce the magnitude of her acceleration in the vertical direction by spreading her arms and increasing air resistance.
- During the dive she can use a power unit strapped to her back to give herself an acceleration in a horizontal direction.
- Alesha's mass, including her equipment, is 100 kg.
- Initially, her position vector is $\begin{pmatrix}-75\\90\\750\end{pmatrix}$ m and her velocity is $\begin{pmatrix}-5\\0\\-10\end{pmatrix}$ m s$^{-1}$.

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(i) Calculate Alesha's initial speed, and the initial angle between her motion and the downward vertical.

**[4]**

At a certain time during the dive, forces of $\begin{pmatrix} 0 \\ 0 \\ -980 \end{pmatrix}$ N, $\begin{pmatrix} 0 \\ 0 \\ 880 \end{pmatrix}$ N and $\begin{pmatrix} 50 \\ -20 \\ 0 \end{pmatrix}$ N are acting on Alesha.

(ii) Suggest how these forces could arise.

**[3]**

(iii) Find Alesha's acceleration at this time, giving your answer in vector form, and show that, correct to 3 significant figures, its magnitude is 1.14 $\mathrm{ms}^{-2}$.

**[3]**

One suggested model for Alesha's motion is that the forces on her are constant throughout the dive from when she leaves the helicopter until she reaches the ground.

(iv) Find expressions for her velocity and position vector at time $t$ seconds after the start of the dive according to this model. Verify that when $t = 30$ she is at the origin.

**[6]**

(v) Explain why consideration of Alesha's landing velocity shows this model to be unrealistic.

**[2]**

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5 The unit vectors **i** and **j** shown in Fig. 2 are in the horizontal andvertically upwards directions.

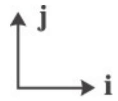


**Fig. 2**

Forces **p** and **q** are given, in newtons, by **p** = 12**i** – 5**j** and **q** = 16**i** + 1.5**j**.

(i) Write down the force **p** + **q** and show that it is parallel to 8**i** – **j**.

**[3]**

(ii) Show that the force 3**p** + 10**q** acts in the horizontal direction.

**[2]**

(iii) A particle is in equilibrium under forces *k***p**, 3**q** and its weight **w**.

Show that the value of *k* must be –4 and find the mass of the particle.

**[3]**



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6 The map of a large area of open land is marked in 1 km squares and a point near the middle of the area is defined to be the origin. The vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ are in the directions east and north.

At time $t$ hours the position vectors of two hikers, Ashok and Kumar, are given by:

Ashok $\quad \mathbf{r}_A = \begin{pmatrix}-2\\0\end{pmatrix} + \begin{pmatrix}8\\1\end{pmatrix}t,$

Kumar $\quad \mathbf{r}_K = \begin{pmatrix}7t\\10-4t\end{pmatrix}.$

(i) Prove that the two hikers meet and give the coordinates of the point where this happens.

[4]

(ii) Compare the speeds of the two hikers.

[3]

7 Two points, A and B, have position vectors **a** = **i** – 3**j** and **b** = 4**i** + 3**j** . The point C lies on the line $y = 1$. The lengths of the line segments AC and BC are equal. Determine the position vector of C. [4]

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8 Fig. 1 shows four forces acting at a point. The forces are in equilibrium.

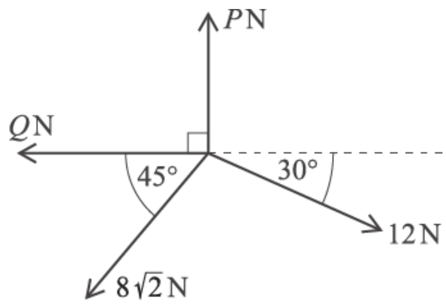


Fig. 1

Show that $P = 14$.

Find $Q$, giving your answer correct to 3 significant figures.

[5]

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9 A model boat has velocity v = $((2t - 2)\mathbf{i} + (2t + 2)\,\mathbf{j})$ m s$^{-1}$ for $t \geq 0$, where $t$ is the time in seconds.
**i** is the unit vector east and **j** is the unit vector north.
When $t = 3$, the position vector of the boat is $(3\mathbf{i} + 14\mathbf{j})$ m.

**(a)** Show that the boat is never instantaneously at rest. **[2]**

**(b)** Determine any times at which the boat is moving directly northwards. **[2]**

**(c)** Determine any times at which the boat is north-east of the origin. **[5]**

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10 Show that points A (1, 4, 9), B (0, 11, 17) and C (3, –10, –7) are collinear. **[4]**

11 OACB is a parallelogram. O is the origin and point A has coordinates (5, 6). Point B has position vector $\mathbf{b} = -2\mathbf{i} - 7\mathbf{j}$.

**(a)** Find the coordinates of point C. **[3]**

M is the midpoint of AB.

**(b)** Prove that $\overrightarrow{OM} = \overrightarrow{MC}$. **[3]**



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**(c)** Find the exact distance MC. **[2]**

12 Points A, B and C have position vectors $-\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ , $4\mathbf{i} - \mathbf{j}$ and $5\mathbf{i} + \mathbf{j} + 5\mathbf{k}$ respectively.

Find the position vector of the point D such that ABCD is a parallelogram. **[4]**

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13 ABCD is a parallelogram. The points A, B and C have position vectors

$$\begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}, \quad \begin{pmatrix} 4 \\ -1 \\ 8 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$$

respectively.

**(a)** Find the position vector of the point D. **[3]**

**(b)** Find the exact distance AC. **[3]**

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14

Point A has position vector $\begin{pmatrix} a \\ b \\ 0 \end{pmatrix}$ where $a$ and $b$ can vary, point B has position vector $\begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}$ and point C has position vector $\begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix}$. ABC is an isosceles triangle with AC = AB.

**(a)** Show that $a - b + 1 = 0$. **[4]**

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(b) Determine the position vector of A such that triangle ABC has minimum area. **[6]**

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15

You are given that $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$ and $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$.

Find a unit vector which is parallel to $\overrightarrow{\mathrm{AB}}$. **[3]**

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16 A circle has centre C (10, 4). The $x$-axis is a tangent to the circle, as shown in Fig. 6.

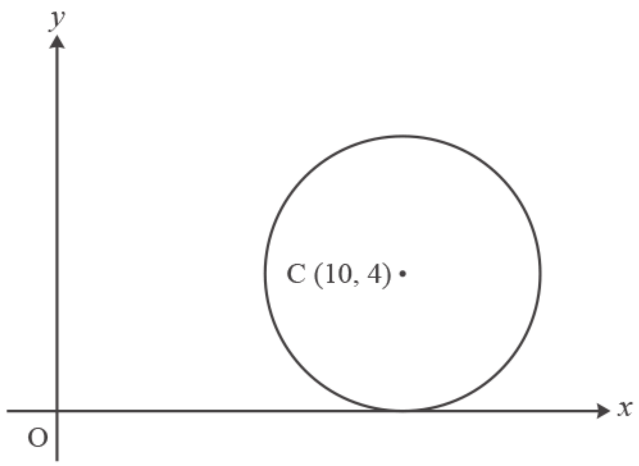


Fig. 6

Write down the position vector of the midpoint of OC. **[1]**

17

The points A and B have position vectors $\mathbf{a} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \\ 8 \end{pmatrix}$ respectively.

Show that the exact value of the distance AB is $\sqrt{101}$ . **[3]**

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18(a) Fig. 15 shows a particle of mass *m* kg on a smooth plane inclined at 30° to the horizontal. Unit vectors **i** and **j** are parallel and perpendicular to the plane, in the directions shown.

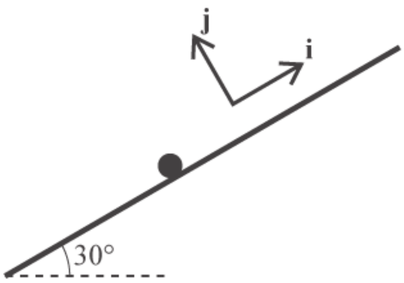


Fig. 15

Express the weight **W** of the particle in terms of $m$, $g$, **i** and **j**. **[2]**

(b) The particle is held in equilibrium by a force **F**, and the normal reaction of the plane on the particle is denoted by **R**. The units for both **F** and **R** are newtons.

Write down an equation relating **W**, **R** and **F**. **[1]**

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(c) Given that $\mathbf{F} = 6\mathbf{i} + 8\mathbf{j}$,

• show that $m = 1.22$ correct to 3 significant figures,

• find the magnitude of $\mathbf{R}$. **[6]**

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19(a)

Fig. 3 shows a triangle PQR. The vector $\overrightarrow{PQ}$ is $\mathbf{i} + 7\mathbf{j}$ and the vector $\overrightarrow{QR}$ is $4\mathbf{i} - 12\mathbf{j}$.

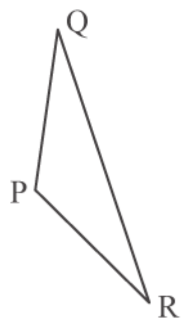


Fig. 3

Show that the triangle PQR is isosceles. **[3]**

(b) The point P has position vector $-3\mathbf{i} - \mathbf{j}$. The point S is added so that PQRS is a parallelogram.

Find the position vector of S. **[2]**

20(a) In this question, the *x*, and y directions are horizontal and vertically upwards respectively.

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A particle of mass 1.5 kg is in equilibrium under the action of its weight and forces $\mathbf{F}_1 = \begin{pmatrix} 4 \\ -2 \end{pmatrix}\text{N}$ and $\mathbf{F}_2$.

Find the force $\mathbf{F}_2$. **[3]**

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(b)

The force $\mathbf{F}_2$ is changed to $\begin{pmatrix} 2 \\ 20 \end{pmatrix}\text{N}$.

Find the acceleration of the particle. **[3]**

21 Fig. 4 shows the regular octagon ABCDEFGH.

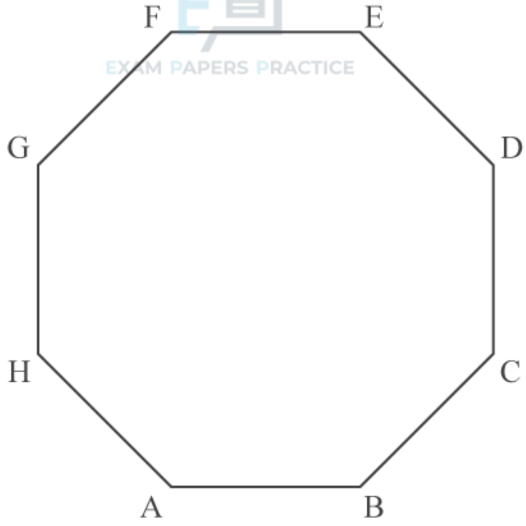


Fig. 4

$\overrightarrow{AB} = \mathbf{i}$, $\overrightarrow{CD} = \mathbf{j}$ where **i** is a unit vector parallel to the *x*-axis and **j** is a unit vector parallel to the *y*-axis.

Find an exact expression for $\overrightarrow{BC}$ in terms of **i** and **j**. [3]

END OF QUESTION PAPER

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| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 1 | | | $\begin{pmatrix}10\\5\\-5\end{pmatrix}=\lambda\begin{pmatrix}3\\2\\-1\end{pmatrix}+\mu\begin{pmatrix}-1\\1\\2\end{pmatrix}$ | M1 | required form, can be soi from two or more correct equations |
| | | | $\Rightarrow 3\lambda - \mu = 10$ | M1 | forming at least two equations and attempting to solve oe |
| | | | $2\lambda + \mu = 5 \Rightarrow 5\lambda = 15,\ \lambda = 3$ | A1 | www |
| | | | $\Rightarrow 9 - \mu = 10,\ \mu = -1$ | A1 | www |
| | | | $-5 = -\lambda + 2\mu,\ -5 = -3 + 2 \times -1$ true | A1 | verifying third equation, **do not** give BOD accept a statement such as $\begin{pmatrix}10\\5\\-5\end{pmatrix}=3\begin{pmatrix}3\\2\\-1\end{pmatrix}+-1\begin{pmatrix}-1\\1\\2\end{pmatrix}$<br>as verification<br>Must **clearly** show that the solutions satisfy all the equations. |
| | | | coplanar | B1 | oe **independent** of all above marks<br><br><u>Examiner's Comments</u><br><br>Most candidates scored the first four marks by forming the equations and solving them.<br><br>Marks were usually lost both when candidates failed to show their solutions worked in all three equations or failed to realise that O, A, B and C must all lie on the same plane for the final mark. |
| | | | Total | 6 | |
| 2 | | i | v = u + a*t* | M1 | May be implied by either of the next two answers but not the final answer. Evidence of use of vectors in question necessary. |
| | | i | Velocity $\mathbf{v}=\begin{pmatrix}2\\0\end{pmatrix}+t\begin{pmatrix}-1\\1\end{pmatrix}$ $\left(=\begin{pmatrix}2-t\\t\end{pmatrix}\right)$ | A1 | |
| | | i | When $t=8$, $\mathbf{v}=\begin{pmatrix}-6\\8\end{pmatrix}$ | A1 | May be implied by the final answer |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | i | speed $\sqrt{(-6)^2+8^2}=10\text{ m s}^{-1}$ | A1 | Cao but condone no units<br>Give SC2 for 10 without working |
| | | ii | $\mathbf{r}=\mathbf{r}_0+\mathbf{u}t+\frac{1}{2}\mathbf{a}t^2$ | 4 | Use of correct equation with substitution. Condone omission of $\mathbf{r}_0$. |
| | | ii | | M1 | Or equivalent equation |
| | | ii | $\mathbf{r}=\begin{pmatrix}0\\-2\end{pmatrix}+\begin{pmatrix}2\\0\end{pmatrix}\times 8+\frac{1}{2}\times\begin{pmatrix}-1\\1\end{pmatrix}\times 8^2$ | A1 | Condone omission of $\mathbf{r}_0$. Follow through for their value of **v** |
| | | ii | $\mathbf{r}=\begin{pmatrix}-16\\30\end{pmatrix}$ | A1 | Cao but may be implied by a correct final answer. |
| | | ii | Distance = 34 m | A1 | Allow for 35.77... from $\mathbf{r}=\begin{pmatrix}-16\\32\end{pmatrix}$ and 37.57... from $\mathbf{r}=\begin{pmatrix}-16\\34\end{pmatrix}$<br><br>**Examiner's Comments**<br><br>This question was about motion in two dimensions using column vectors. It was well answered. Such marks as were lost were usually as a result of candidates not fully answering the questions, omitting the velocity at time t and the speed in part (i) and the distance travelled in part (ii). |
| | | | Total | 8 | |

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| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 3 | | i | **p** $\sqrt{(-1)^2+(-1)^2+5^2}=\sqrt{27}$<br>**q** $\sqrt{(-1)^2+(-4)^2+2^2}=\sqrt{21}$<br>**r** $\sqrt{2^2+5^2+0^2}=\sqrt{29}$ | M1 | Use of Pythagoras<br><br>Note Magnitudes are 5.196, 4.583 and 5.385 respectively |
| | | i | Greatest magnitude: **r** | A1 | <u>**Examiner's Comments**</u><br><br>Candidates were asked to find which of three forces, given as 3-dimensional column vectors, had the greatest magnitude. Almost all candidates got this right. |
| | | ii | Weight $=\begin{pmatrix}0\\0\\-4\end{pmatrix}$ | B1 | Condone $g$ = 9.8 giving weight is<br><br>$\begin{pmatrix}0\\0\\-3.92\end{pmatrix}$ N. Accept 4↓. |
| | | ii | $\mathbf{p}+\mathbf{q}+\mathbf{r}+\mathbf{weight}=\begin{pmatrix}0\\0\\3\end{pmatrix}$ | | $g$ = 9.8 gives $\begin{pmatrix}0\\0\\3.08\end{pmatrix}$ |
| | | ii | $0.4\mathbf{a}=\begin{pmatrix}0\\0\\3\end{pmatrix}$ | B1 | Relevant attempt at Newton's 2nd Law. The total force must be expressed as a vector in some form. For this mark allow the weight to be missing, in the wrong component or to have the wrong sign. Condone *mg* in place of *m* for this mark only. |
| | | ii | Magnitude of acceleration is 7.5 ms$^{-2}$ | B1 | CAO apart from using $g$ = 9.8 ⇒ $a$ = 7.7 |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | ii | Direction is vertically upwards | B1 | **Examiner's Comments**<br><br>This question was about the application of Newton's second law to an object subject to the same three forces and its weight. Candidates needed to write the weight of the object in vector form. Many candidates got this completely right but others made mistakes with the weight, some applying it in the wrong direction or all three directions. |
| | | | Total | 6 | |

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| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 4 | | i | Speed = $\sqrt{(-5)^2+0^2+(-10)^2}$ | M1 | For use of Pythagoras. Accept $\sqrt{5^2+10^2}$ . |
| | | i | = 11.2 ms$^{-1}$ (11.18) | A1 | Accept $\sqrt{125}$ or $5\sqrt{5}$ |
| | | i | $\tan\theta=\frac{5}{10}$ | M1 | Complete method for correct angle; may use $\sin\theta=\frac{5}{11.2}$, $\cos\theta=\frac{10}{11.2}$. |
| | | i | α = 26.6° | A1 | Allow 153.4°, 206.6°<br><br>**Examiner's Comments**<br><br>This question was about motion in three dimensions described using vectors. It was also about modelling. The context was a sky-dive. Candidates were asked to use the given (vector) initial velocity to find the initial speed and direction. Nearly all candidates found the speed but there were many mistakes with the direction, particularly finding the wrong angle or using the initial displacement instead of the initial velocity. |
| | | ii | $\begin{pmatrix}0\\0\\-980\end{pmatrix}$ her weight | B1 | The descriptions should be linked to the forces, either by the layout of the answer or by suitable text. If not, assume that the forces they refer to are in the order given here (which is the same as the question). |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | ii | $\begin{pmatrix} 0 \\ 0 \\ 880 \end{pmatrix}$ resistance to her vertical motion | B1 | Accept “Air resistance”, “Arms stretched out” and similar statements.<br>Condone mention of a parachute.<br><br>**Examiner's Comments**<br><br>This question was about motion in three dimensions described using vectors. It was also about modelling. The context was a sky-dive. Candidates were asked to account for the three forces acting on the sky-diver. This involved interpreting the information given in the question. Many did this correctly but there were also candidates who dropped marks here. |
| | | ii | $\begin{pmatrix} 50 \\ -20 \\ 0 \end{pmatrix}$ force from the power unit | B1 | |
| | | iii | Resultant force $= \begin{pmatrix} 50 \\ -20 \\ -100 \end{pmatrix}$ | B1 | May be implied |
| | | iii | Acceleration $= \begin{pmatrix} 0.5 \\ -0.2 \\ -1 \end{pmatrix}$ | B1 | Newton's 2nd Law |
| | | iii | Magnitude $= \sqrt{0.5^2 + (-0.2)^2 + 1^2} = 1.1357...$ | | |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | iii | So 1.14 to 3 s.f. | B1 | Answer given. Allow FT from sign errors.<br>Accept \|**F**\| ÷ 100<br><br>**Examiner's Comments**<br><br>This question was about motion in three dimensions described using vectors. It was also about modelling. The context was a sky-dive. Candidates were asked to find the acceleration of the sky-diver as a vector and to verify its magnitude. While this was on the whole well answered, a number of candidates omitted to find the vector form and so lost a mark. |
| | | iv | **v** = **u** + **a***t* | M1 | FT their **a** for the first 5 marks of this part. Vectors must be seen or implied. Accept valid integration. |
| | | iv | $\mathbf{v} = \begin{pmatrix} -5 \\ 0 \\ -10 \end{pmatrix} + \begin{pmatrix} 0.5 \\ -0.2 \\ -1 \end{pmatrix} t$ | A1 | |
| | | iv | $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$ | M1 | Vectors must be seen or implied. Accept valid integration.<br>Condone no $r_0$ for this M mark |
| | | iv | $\mathbf{r} = \begin{pmatrix} -75 \\ 90 \\ 750 \end{pmatrix} + \begin{pmatrix} -5 \\ 0 \\ -10 \end{pmatrix} t + \frac{1}{2}\begin{pmatrix} 0.5 \\ -0.2 \\ -1 \end{pmatrix} t^2$ | A1 | |
| | | iv | When *t* = 30 | M1 | Vectors must be seen or implied. |
| | | iv | $\mathbf{r} = \begin{pmatrix} -75 - 150 + 225 \\ 90 + 0 - 90 \\ 750 - 300 - 450 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, as required | E1 | CAO |

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| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | iv | | | SC1 to replace the first 4 marks of this section if the acceleration is taken to be **g** but the answer is otherwise correct.<br><br>**Examiner's Comments**<br><br>This question was about motion in three dimensions described using vectors. It was also about modelling. The context was a sky-dive. Carried 6 marks and many candidates obtained all of them. Candidates were asked to find expressions for the velocity and position vector at time t, and then to show that at a given time the sky-diver was at the origin. The commonest mistake among those who otherwise answered this part well was to omit the initial displacement from the position vector. However, several candidates used g for the acceleration instead of the acceleration they should have found in part (iii). |
| | | v | When $t = 30$, $\mathbf{v} = \begin{pmatrix} 10 \\ -6 \\ -40 \end{pmatrix}$ | M1 | There must be an attempt to work out at least the vertical component of the velocity at $t = 30$. This mark is not dependent on a correct answer. |
| | | v | The vertical component of the velocity is too fast for a safe landing | A1 | Accept an argument based on speed derived from a vector.<br><br>**Examiner's Comments**<br><br>This question was about motion in three dimensions described using vectors. It was also about modelling. The context was a sky-dive. Was not well answered. Candidates were asked to show that a consideration of the sky-diver's landing velocity showed the model to be unrealistic. This was part of the overall modelling process; they had considered the landing position in part (iv) and were now required to consider the velocity on landing. So they needed to calculate the velocity on landing, or at least its vertical component. Most candidates failed to do this and made comments that were not based on evidence obtained from the question. |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | | Total | 18 | |

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| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 5 | | i | p + q = 28i – 3.5j | B1 | |
| | | i | 28i – 3.5j = $k$(8i – j) | M1 | Or equivalent. $k$ may be implied by going straight to 3.5 |
| | | i | $k$ = 3.5 | A1 | |
| | | | (So they are parallel) | | |
| | | | Alternative | | |
| | | i | p + q = 28i – 3.5j | B1 | |
| | | | $\mathbf{p}+\mathbf{q}$: $\tan\theta = \frac{-3.5}{28} \Rightarrow \theta = -7.13°$ | | |
| | | i | $8\mathbf{i}-\mathbf{j}$: $\tan\theta = \frac{-1}{8} \Rightarrow \theta = -7.13°$ | M1 | Comparing the ratio of the components in each of the two vectors is sufficient, using any consistent sign convention. The angle does not need to be worked out, nor does tan have to be seen. |
| | | i | So they are parallel | A1 | Both ratios the same and correct |
| | | ii | 3p + 10q = (36 + 160)i + (–15 + 15)j | | |
| | | ii | =196i | B1 | |
| | | ii | Zero j component so horizontal | B1 | Or equivalent explanation. May be shown on a diagram |
| | | iii | The horizontal component must be zero | | Substituting $k$ = –4 and showing **i** component is zero is acceptable<br>Award for 24.5 seen<br>Award for 2.5 seen. FT from their weight. |
| | | iii | So 12$k$ + 3 × 16 = 0 ⇒ $k$ = –4 | B1 | Substituting $k$ = –4 and showing **i** component is zero is acceptable |
| | | iii | w = –24.5j | B1 | Award for 24.5 seen |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | iii | $mg = 24.5 \Rightarrow m = 2.5$ The mass is 2.5 kg | B1 | Award for 2.5 seen. FT from their weight.<br><br>**Examiner's Comments**<br><br>This question was about vectors defined using i, j notation. It was well answered with many candidates obtaining full marks. Most of the errors that did occur were in part (iii) where candidates were required to interpret the vectors as forces on a particle in equilibrium, with its weight now introduced. Some did not interpret the equilibrium conditions sufficiently rigorously, failing to recognise that both the horizontal and vertical components of the resultant had to be zero. The question ended with a request for the mass of the particle and some candidates confused this with its weight. |
| | | | Total | 8 | |

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| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 6 | | i | Either $-2 + 8t = 7t$<br>Or $t = 10 - 4t$ | M1 | Forming an equation for $t$. Accept vector equation for this mark. May be implied by a statement that $t = 2$. |
| | | i | $\Rightarrow t = 2$ | A1 | |
| | | i | Substituting $t = 2$ in **both** expressions | B1 | oe, eg showing $t = 2$ satisfies both equations or a vector equation. |
| | | i | They meet at (14, 2) | B1 | Accept $\begin{pmatrix} 14 \\ 2 \end{pmatrix}$<br><br>**Examiner's Comments**<br><br>Candidates were asked to prove the two hikers meet and this involved showing their position vectors are the same at some time ($t = 2$). Many lost a mark by not showing that this was true for both components when the position vectors were equated. |
| | | ii | Ashok's speed is $\sqrt{8^2 + 1^2} = \sqrt{65}$ | B1 | |
| | | ii | Kumar's speed is<br><br>$\sqrt{7^2 + (-4)^2} = \sqrt{65}$ km h$^{-1}$ | B1 | |
| | | ii | They both walk at the same speed | B1 | CAO from correct speeds<br><br>SC1 for finding both velocities correctly but neither speed<br><br>**Examiner's Comments**<br><br>Candidates were asked to compare the speeds of the two hikers and most got this right. A few compared their velocities. Others did not know how to obtain the velocities from the given expressions in terms of $t$ for the position vectors. |
| | | | Total | 7 | |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 7 | | | Midpoint of $AB = \left(\frac{5}{2}, 0\right)$ | M1(AO3.1a) | |
| | | | Gradient of perpendicular to $AB = -\frac{1}{2}$ | M1(AO1.1) | |
| | | | Perpendicular bisector equation is $y = -\frac{1}{2}\left(x - \frac{5}{2}\right)$oe | A1(AO1.1) | |
| | | | Position vector is $\frac{1}{2}\mathbf{i} + \mathbf{j}$ | A1(AO1.1) | |
| | | | | [4] | |
| | | | **Alternative method** | | |
| | | | Suppose C has position vector $\mathbf{c} = p\mathbf{i} + \mathbf{j}$<br>$\overrightarrow{\mathbf{AC}}$ = ( $p$ –1)**i** + 4**j** oe or $\lvert AC\rvert^2 = (p-1)^2 + 4^2$ oe | M1(AO3.1a) | |
| | | | $\overrightarrow{\mathbf{BC}}$ = ( $p$ – 4)**i** – 2**j** oe or $\lvert BC\rvert^2 = (p-4)^2 + 2^2$ oe | M1(AO1.1) | |
| | | | $(p-1)^2 + 4^2 = (p-4)^2 + 2^2$ | A1(AO1.1) | soi |
| | | | Position vector is $\frac{1}{2}\mathbf{i} + \mathbf{j}$ | A1(AO1.1) | |
| | | | Total | 4 | |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 8 | | | $P = 8\sqrt{2}\sin 45° + 12\sin 30°$ | M1<br>M1 | Considering equilibrium in the vertical direction<br>Resolution of forces of 12 N and $8\sqrt{2}$ N in the vertical direction.<br>Do not allow sin-cos interchange for the 30° angle. |
| | | | *P* = 14 | A1 | Dependent on both M marks |
| | | | $Q + 8\sqrt{2}\cos 45° = 12\cos 30°$ | B1 | |
| | | | *Q* = 2.39 | B1 | **Examiner's Comments**<br>This question was answered correctly by almost all candidates. A small number made sign errors, particularly when finding *Q*. There were also those who did not give *Q* to 3 significant figures, as requested in the question. |
| | | | Total | 5 | |

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| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 9 | | a | Require both components zero at the same *time* **i** component zero only when $t = 1$ and **j** component only when<br><br>$t = -1$ so there are no such times | M1(AO3.1b)<br><br>A1(AO2.4)<br><br>[2] | May be implied but must be clear<br><br>Or say **j** component $\geq 2$ since $t \geq 0$ |
| | | b | This requires use of the velocity vector<br>Travelling due north means that the **i** component is zero and the **j** component +ve<br>So we need $2t - 2 = 0$ for **i** component, giving $t = 1$.<br>This gives **j** component $4 > 0$ so yes at $t = 1$. | M1(AO3.3)<br><br>A1(AO2.4)<br><br>[2] | Recognise velocity vector required<br><br>Must test **j** component |
| | | c | This requires use of the position vector<br>**either**<br><br>$\mathbf{r} = \int \mathbf{v}\,dt$ so $\mathbf{r} = \int((2t-2)\mathbf{i} + (2t+2)\mathbf{j})dt = (t^2 - 2t + C)\mathbf{i} + (t^2 + 2t + D)\mathbf{j}$<br><br>$\mathbf{r} = 3\mathbf{i} + 14\mathbf{j}$ when $t = 3$ so $C = 0$ and $D = -1$ so $\mathbf{r} = (t^2 - 2t)\mathbf{i} + (t^2 + 2t - 1)\mathbf{j}$<br>Or<br>$\mathbf{a} = 2\mathbf{i} + 2\mathbf{j}$ when $t = 3$ $\mathbf{v} = 4\mathbf{i} + 8\mathbf{j}$ ,<br>$\mathbf{r} = (4\mathbf{i} + 8\mathbf{j})(t-3) + \frac{1}{2}(2\mathbf{i} + 2\mathbf{j})(t-3)^2 + 3\mathbf{i} + 14\mathbf{j}$<br><br>and so $\mathbf{r} = (t^2 - 2t)\mathbf{i} + (t^2 + 2t - 1)\mathbf{j}$<br><br>the boat is NE of O when the **i and j** components are equal and +ve<br><br>we require $t^2 - 2t = t^2 + 2t - 1$ so $t = 0.25$ this gives components of $-0.4375$ so no. | M1(AO3.1b)<br><br>M1(AO1.1)<br><br>A1(AO1.1)<br><br>M1(AO1.1)<br>A1(AO1.1)<br><br>M1(AO3.2b)<br><br>A1(AO2.1)<br><br>[5] | Recognise position vector required<br><br>May use + **C** instead<br><br>Must find **a** but may omit 3**i** + 14**j**<br><br>Award even if +ve not mentioned<br><br>Must be complete argument |
| | | | Total | 9 | |

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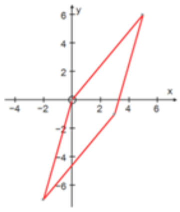
| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 10 | | | $\overrightarrow{AB}=\begin{pmatrix}-1\\7\\8\end{pmatrix}$ | M1(AO3.1a) | Attempt to find vector between any two of the points |
| | | | $\overrightarrow{AC}=\begin{pmatrix}2\\-14\\-16\end{pmatrix}$ | A1(AO1.1) | Correct pair of vectors with common point $\overrightarrow{BC}=\begin{pmatrix}3\\-21\\-24\end{pmatrix}$ |
| | | | AB is parallel to AC | B1(AO1.1) | |
| | | | Common point A so collinear | E1(AO2.1) | |
| | | | | [4] | |
| | | | Total | 4 | |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 11 | | a | <br>a = 5i + 6j<br>$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = \overrightarrow{OB} + \overrightarrow{OA} = \mathbf{b} + \mathbf{a}$<br>$= (5\mathbf{i} + 6\mathbf{j}) + (-2\mathbf{i} - 7\mathbf{j}) = 3\mathbf{i} - \mathbf{j}$<br>So C is (3, –1) | M1(AO 1.1a)<br>M1(AO 3.1a)<br>E1(AO 3.2a)<br>[3] | Using vector notation<br>Using **a** for vector $\overrightarrow{BC}$ in an addition<br>Must be coordinates; do not allow for position vector only |
| | | b | $\overrightarrow{OM} = \frac{1}{2}(\mathbf{a}+\mathbf{b}) = \frac{3}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}$<br>$\overrightarrow{MC} = (3\mathbf{i} - \mathbf{j}) - \left(\frac{3}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}\right)$<br>$= \left(\frac{3}{2}\mathbf{i} - \frac{1}{2}\mathbf{j}\right) = \overrightarrow{OM}$ | B1(AO 2.1)<br>M1(AO 1.1a)<br>A1(AO 2.1)<br>[3] | |
| | | c | $\lvert\overrightarrow{MC}\rvert = \left\lvert \frac{3}{2}\mathbf{i} - \frac{1}{2}\mathbf{j} \right\rvert = \frac{1}{2}\sqrt{3^2 + (-1)^2}$<br>$= \frac{1}{2}\sqrt{10}$ | M1(AO 1.1a)<br>A1(AO 1.1b)<br>[2] | Or exact equivalent, e.g. $\sqrt{\left(\frac{3}{2}\right)^2 + \left(-\frac{1}{2}\right)^2}$<br>Accept alternative answer $\sqrt{\frac{5}{2}}$ |
| | | | Total | 8 | |

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| Question | | | Answer/Indicative content | Marks | Guidance | |
|---|---|---|---|---|---|---|
| 12 | | | $\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}$<br><br>$=5\mathbf{i}-4\mathbf{j}-2\mathbf{k}$<br><br>$\overrightarrow{DC}=5\mathbf{i}-4\mathbf{j}-2\mathbf{k}$ oe<br><br>$\overrightarrow{OD}=5\mathbf{j}+7\mathbf{k}$ | M1(AO 1.1)<br><br>A1(AO 1.1)<br><br>M1(AO 3.1a)<br><br>A1(AO 2.2a)<br><br>[4] | Attempt to subtract two position vectors to get the vector for a side<br><br>Equating opposite side to vector found **or** finding $\overrightarrow{AC}$ and using $\overrightarrow{AB}+\overrightarrow{AD}=\overrightarrow{AC}$ oe | Allow alternative vector notation<br><br>Allow $\overrightarrow{CD}$□□here, but no ft from $\overrightarrow{CD}$ for final answer |
| | | | Total | 4 | | |
| 13 | | a | $\overrightarrow{OD}=\overrightarrow{OA}+\overrightarrow{AD}$<br><br>$\overrightarrow{AD}=\overrightarrow{BC}=\overrightarrow{OC}-\overrightarrow{OB}$<br><br>$\overrightarrow{OD}=\begin{pmatrix}3\\1\\-2\end{pmatrix}+\begin{pmatrix}0\\2\\5\end{pmatrix}-\begin{pmatrix}4\\-1\\8\end{pmatrix}=\begin{pmatrix}-1\\4\\-5\end{pmatrix}$ | M1(AO3. 1a)<br><br>M1(AO3. 1a)<br><br>A1(AO3. 1b)<br><br>[3] | Use of $\|\|^{gm}$ and vector subtraction soi | $\overrightarrow{OD}=\overrightarrow{OA}+\overrightarrow{AD}$<br><br>$\overrightarrow{AD}=\overrightarrow{BC}=\overrightarrow{OC}-\overrightarrow{OB}$ |
| | | b | $\overrightarrow{AC}=\begin{pmatrix}-3\\1\\7\end{pmatrix}$<br><br>$AC=\sqrt{9+1+49}$<br>$=\sqrt{59}$ | B1(AO1. 1a)<br><br>M1(AO1. 1a)<br>A1(AO1. 1b)<br><br>[3] | | Or $\overrightarrow{CA}=\begin{pmatrix}3\\-1\\-7\end{pmatrix}$ |
| | | | Total | 6 | | |

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| Question | | | Answer/Indicative content | Marks | Guidance | |
|---|---|---|---|---|---|---|
| 14 | | a | $\overrightarrow{AC} = \begin{pmatrix} 2-a \\ 4-b \\ 2 \end{pmatrix}$, $\overrightarrow{AB} = \begin{pmatrix} 4-a \\ 2-b \\ 0 \end{pmatrix}$ | M1 (AO 1.1) | Forming vectors for sides AB and AC | Implied by next M1 |
| | | | $(4 - a)^2 + (2 - b)^2 = (2 - a)^2 + (4 - b)^2 + 4$ o.e. | M1 (AO 1.1a) | Use of AB = AC | |
| | | | $16 - 8a + a^2 + 4 - 4b + b^2 = 4 - 4a + a^2 + 16 - 8b + b^2 + 4$ | M1 (AO 1.1) | expanding | |
| | | | $4a - 4b + 4 = 0 \Rightarrow a - b + 1 = 0$ | A1 (AO 2.1) | AG Convincing completion | |
| | | | | [4] | <u>Examiner's Comments</u><br>This part was generally answered well. | |

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| Question | | | Answer/Indicative content | Marks | Guidance | |
|---|---|---|---|---|---|---|
| | | b | D has position vector [illegible] where D is midpoint of BC | B1 (AO 3.1a) | Midpoint<br><br>OR if clearly minimising AC or AB – **M1** for relevant vector using a and b (May be implied by second M1) | |
| | | | $\overrightarrow{AD}=\begin{pmatrix}3-a\\2-a\\1\end{pmatrix}$ | M1 (AO 1.1) | Finding relevant vector in terms of a or b only | |
| | | | $\text{Area}=\frac{1}{2}AD.BC=\frac{2\sqrt{3}\sqrt{(3-a)^2+(2-a)^2+1}}{2}$ | M1(AO 1.1) | Expression for AD or $AD^2$ (correct method but may have errors) | May use area proportional to AD, AC or AB without calculation of expression for area |
| | | | $\sqrt{3}\sqrt{2\left((a-2.5)^2+0.75\right)}$ | M1(AO 3.1a) | Completion of square | Or differentiation of AD, $AD^2$, AC, AB, $AC^2$ or $AB^2$ |
| | | | *a* = 2.5 for min | A1(AO 2.2a) | | |
| | | | Position vector $\begin{pmatrix}2.5\\3.5\\0\end{pmatrix}$ | A1(AO 3.2a)<br><br>[6] | <u>**Examiner's Comments**</u><br><br>This non-standard problem led to many different approaches but finding the midpoint of BC seemed the most logical start. The need to find a relevant vector such as A to the midpoint of *BC* in terms of one variable was appreciated by most candidates. The challenging part of this question was to find a for the minimum area and then use it to find the correct vector. | |
| | | | Total | 10 | | |

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| Question | | | Answer/Indicative content | Marks | Guidance | |
|---|---|---|---|---|---|---|
| 15 | | | b – $a$ = 6i – 6j – 3k | M1 (AO1.1) | allow one slip | or in column vector form |
| | | | $\lvert AB \rvert = \sqrt{6^2 + (-6)^2 + (-3)^2}$ | M1 (AO1.1) | FT their **b** – **a** | allow use of 2i – 2j – **k** oe |
| | | | ⅔i – ⅔j – ⅓K | A1 (AO1.1) [3] | accept eg $^6/_9$**i** – $^6/_9$**j** – $^3/_9$**k** | or in column vector form |
| | | | Total | 3 | | |
| 16 | | | 5i + 2j or $\begin{pmatrix} 5 \\ 2 \end{pmatrix}$ | B1 (AO2.5) [1] | Must be correct vector notation; not (5, 2)<br><br>__Examiner's Comments__<br><br>In this part nearly all could find the midpoint but it was often left as the coordinate (5, 2) rather than given in vector form. | |
| | | | Total | 1 | | |
| 17 | | | $\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} -4 \\ 2 \\ 9 \end{pmatrix}$ | | AG | |
| | | | | B1(AO2.1) | Soi Allow for $\overrightarrow{BA}$ even if wrongly labelled | |
| | | | $\text{distance} = \sqrt{(-4)^2 + 2^2 + 9^2} = \sqrt{101}$ | M1(AO2.1) | Attempt to find magnitude of their displacement vector | Do not allow final mark when there are missing brackets eg $-4^2 + 2^2 + 9^2$ |
| | | | | A1(AO2.1) [3] | cao AG | |
| | | | Total | 3 | | |

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| Question | | | Answer/Indicative content | Marks | Guidance | |
|---|---|---|---|---|---|---|
| 18 | a | | **W** = (–*mg* sin 30°)**i** + (–*mg* cos 30°)**j**<br>$\left[\mathbf{W}=\left(-\frac{1}{2}mg\right)\mathbf{i}+\left(-\frac{\sqrt{3}}{2}mg\right)\mathbf{j}\right]$ | M1(AO3.1b)<br>A1(AO2.5)<br>[2] | Attempting to resolve the weight. Allow sin / cos interchange and sign errors for the method mark.<br>All correct in this vector form | *mg* must be seen for the method mark. |
| | b | | **W** + **R** + **F** = 0 | B1(AO2.5)<br>[1] | Allow any rearrangement of this. Allow if their expression for **W** is used instead of **W** | |
| | c | | **R** = *R***j**<br><br>[(-*mg* sin 30°) **i** + (-*mg* cos30°) **j** + (6**i** + 8**j**) + *R***j** = **0**]<br><br>**i** component<br>–*mg* sin 30° +6 = 0<br><br>giving *m* = 1.22 to 3 sf<br><br>**j** component: –12cos 30 + 8 + *R* = 0<br><br>*R* = 2.39 so magnitude is 2.39 N | B1(AO3.1b)<br><br>M1(AO3.1a)<br><br>A1(AO1.1)<br><br>A1(AO1.1)<br><br>M1(AO3.1a)<br><br>A1(AO3.2a)<br>[6] | Allow for any clear indication that **R** is a multiple of **j** or that it has no component in the **i** direction<br><br>Forming equation from their **i** terms, or equivalent by resolving parallel to the plane. FT their **W** Correct equation in **i** direction **AG**<br><br>Equation from the **j** terms (must include all three terms), oe, and using value of *m* Accept arwt 2. 4 | May be implied with an equation for the i direction with two terms and an equation in the j direction with three terms<br><br>12 is the value for *mg.* 1.22 × 9.8 = 11.956 |
| | | | Total | 9 | | |

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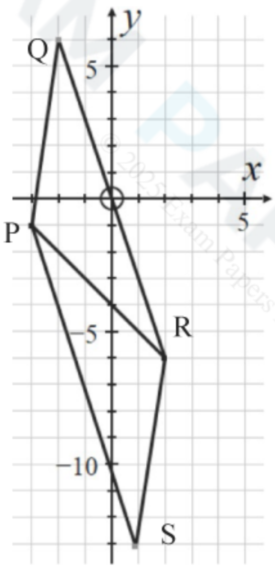
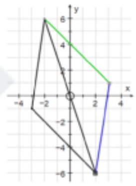
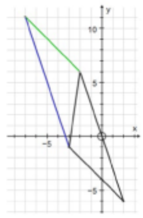
| Question | | | Answer/Indicative content | Marks | Guidance | |
|---|---|---|---|---|---|---|
| 19 | a | | $\left\lvert\overrightarrow{PQ}\right\rvert = \sqrt{1^2+7^2} = \sqrt{50}$<br>$\overrightarrow{PR} = (\mathbf{i}+7\mathbf{j})+(4\mathbf{i}-12\mathbf{j}) = 5\mathbf{i}-5\mathbf{j}$<br>$\left\lvert\overrightarrow{PR}\right\rvert = \sqrt{5^2+5^2} = \sqrt{50}$<br>So the triangle is isosceles | B1(AO1.1a)<br>M1(AO1.1a)<br><br>A1<br>(AO2.2a)<br><br>[3] | Allow for $PQ^2$<br><br>Attempt to add vectors<br><br>Must deduce the triangle is isosceles from correct working | Allow finding $\overrightarrow{RP} = -5\mathbf{i}+5\mathbf{j}$ |
| | b | | PQRS parallelogram so $\overrightarrow{PS} = \overrightarrow{QR} = 4\mathbf{i}-12\mathbf{j}$<br><br>Position vector<br><br>$\overrightarrow{OS} = \overrightarrow{OP}+\overrightarrow{PS} = (-3\mathbf{i}-\mathbf{j})+(4\mathbf{i}-12\mathbf{j}) = \mathbf{i}-13\mathbf{j}$<br> | M1(AO3.1a)<br><br>A1(AO1.1)<br><br>[2] | Using the properties of the parallelogram<br><br>cao | SPECIAL CASES<br><br>Allow **SC1** for correct answer for either<br>PQSR or PSQR<br>If PQSR used, then<br>$\overrightarrow{OS} = 3\mathbf{i}+\mathbf{j}$<br><br>If PSQR used, then<br>$\overrightarrow{OS} = -7\mathbf{i}+11\mathbf{j}$<br> |
| | | | Total | 6 | | |

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| Question | | | Answer/Indicative content | Marks | Guidance | |
|---|---|---|---|---|---|---|
| 20 | a | | weight $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix}$<br><br>Equilibrium equation<br>$\begin{pmatrix} 0 \\ -1.5g \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \mathbf{F}_2 = \mathbf{0}$<br>$\mathbf{F}_2 = \begin{pmatrix} -4 \\ 2+1.5g \end{pmatrix} = \begin{pmatrix} -4 \\ 16.7 \end{pmatrix}$ N | B1(AO1.2)<br><br>M1(AO1.1a)<br><br>A1(AO1.1)<br>[3] | Allow seen or implied by correct answer<br><br>**FT** their weight. Allow sign errors<br><br>must be vector allow **i-j** form | |
| | b | | Newton's second law<br>$\begin{pmatrix} 0 \\ -1.5g \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ 20 \end{pmatrix} = m\mathbf{a}$<br>$\mathbf{a} = \frac{1}{1.5}\begin{pmatrix} 6 \\ 3.3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2.2 \end{pmatrix}$ m s$^{-2}$ | M1(AO1.1a)<br><br>A1(AO1.1)<br>[2] | Addition of vectors. Allow if weight missing but other two forces and acceleration seen in equation<br>Must be vector; any correct form. | Allow wrong weight only if given as a vector |
| | | | Total | 4 | | |
| 21 | | | BC is parallel to $\boldsymbol{i} + \boldsymbol{j}$ OR 1 unit at 45°<br><br>$\lvert\boldsymbol{i} + \boldsymbol{j}\rvert = \sqrt{2}$<br><br>$\overrightarrow{BC} = \frac{1}{\sqrt{2}}(\boldsymbol{i} + \boldsymbol{j})$ oe | M1 (AO3.1a)<br>M1 (AO1.1)<br>A1 (AO2.2a)<br>[3] | Eg k**i** + k**j**<br><br>Could be on diagram<br><br>Must be exact eg cos45**i** + sin45**j**<br><br>__Examiner's Comments__<br><br>In this question although most realised that BC was parallel to **i** + **j**, only the most able realised they had to consider the magnitude of that vector sum. | |
| | | | Total | 3 | | |

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